

KOLEJ UNIVERSITI TUNKU ABDUL RAHMAN
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2020/2021

APRIL/MAY EXAMINATION

COMPUTER SCIENCE BACS2163
SOFTWARE ENGINEERING

SATURDAY, 8 MAY 2021

TIME: 2.00 PM – 5.00 PM (3 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

Instructions to Candidates:

Answer **ALL** questions in the requested format or template provided.

- This is an open book final online assessment. You **MUST** answer the assessment questions on your own without any assistance from other persons.
- You must submit your answers within the following time frame allowed for this online assessment:
 - The deadline for the submission of your answers is **half an hour** from the end time of this online assessment.
- Penalty as below **WILL BE IMPOSED** on students who submit their answers late as follows:
 - The final marks of this online assessment will be reduced by 10 marks for answer scripts that are submitted within 30 minutes after the deadline for the submission of answers for this online assessment.
 - The final marks of this online assessment will be downgraded to zero (0) mark for any answer scripts that are submitted after one hour from the end time of this online assessment.
- Extenuation Mitigating Circumstance (EMC) encountered, if any, must be submitted to the Faculty/Branch/Centre within 48 hours after the date of this online assessment. All EMC applications must be supported with valid reasons and evidence. The UC EMC Guidelines apply.

FOCS Additional Instructions to Candidates:

- Include your **FULL NAME, STUDENT ID** and **PROGRAMME OF STUDY** in your submission of answer.
- Read all the questions carefully and understand what you are being asked to answer.
- Marks are awarded for your own (original) analysis. Therefore, use the time and information to build well-constructed answers.

BACS2163 SOFTWARE ENGINEERING**STUDENT'S DECLARATION OF ORIGINALITY**

By submitting this online assessment, I declare that this submitted work is free from all forms of plagiarism and for all intents and purposes is my own properly derived work. I understand that I have to bear the consequences if I fail to do so.

Final Online Assessment Submission

Course Code:

Course Title:

Signature:

Name of Student:

Student ID:

Date:

BACS2163 SOFTWARE ENGINEERING**Question 1**

Online ordering has becoming popular in Malaysia due to COVID-19 during Movement Control Order (MCO). There are about 68% of population who are actively involved in online ordering activities to get products from both local and overseas stores. Foreseeing the online ordering market is expanding in near future, CBB company is planning to develop an *online Enterprise Resource Planning (ERP) management system* to ease local store sellers to manage their products upload, order management, delivery tracking, finance reports, and connection with China suppliers.

CBB online ERP system is a platform that will link local Malaysia seller to get products from China suppliers and sell in Malaysia online stores. Functions in the platform including: get product pictures from China suppliers' website automatically and upload into Shopee, Lazada and 11Street, transfer order from Malaysia buyers to China suppliers, edit the products information, calculate shipping fee automatically, process return/refund, process international shipping and etc. CBB management has decided to integrate their existing warehouse system into this new online ERP system for buyers, sellers and suppliers to keep track of their products. Therefore, the new ERP system shall be confirmed to ISO 9001:2008 as practised by the existing warehouse system.

The existing CBB warehouse system is developed in 1998 and have been re-engineered once in 2008 which focus in reverse engineering (recovering documentation). Although the system is developed in .NET platform, the structure of the system is yet to be improved. Since then, IT department has been planning to continue second re-engineering phase in order to accomplish a full re-engineering cycle. This will ease the maintenance of CBB warehouse system.

Since this online ERP system is integrating both Malaysia and China, the management has assigned a lawyer in consulting legislation issues such as general and special goods specification, illegal shipping, shipping charges based on different types of goods, declaration of goods, and etc. Besides that, an internal consultant is also invited to join the project to handle matters regarding Privacy, Safety and Ethical issues in e-Commerce. A full training regarding the ERP system's functions, rules and regulations, will be provided to all ERP system's suppliers and sellers as this system might be complicated to use. This is to ensure the users are fully equipped with the necessary knowledge in using the system.

CBB has assigned this project to their IT department with 4 senior software engineers and lead by you, a senior IT manager. IT departments have been developing all CBB internal systems (including the existing warehouse system). You have been given sufficient budget with first release of the online ERP system to the market within 1.5 month. Subsequent release must be planned within 1–2-month timeframe. All development must be done within the IT department without recruiting additional software engineers. The whole project shall be kept within four months.

- a) Identify and explain **THREE (3)** software quality attributes for the CBB online ERP system. (9 marks)
- b) Based on the case study, which Software Process Model would be suitable for this project? Explain your answer based on the **FOUR (4)** criteria. (10 marks)
- c) Software Engineering is a layered technology consisting of Quality, Process, Methods, and Tools. Explain **ONE (1)** appropriate Method and **TWO (2)** Tools (based on Functional Classification of Computer Aided Software Engineering CASE technology) that can support Component Based Software Engineer (CBSE). Explain your answer. (6 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 2**

Please answer the following questions based on **CBB case study in Question 1**.

- a) Assuming that the partial schedule planning is shown in the following table. Complete the planning by adding **ONE (1)** cycle: Cycle 4, with all the processes remaining the same and the whole project schedule shall be kept within four months. Illustrate a Gantt Chart for the project with four cycles. Project will start in June 2021.

(2 + 9 marks)

Activity	Duration (month)	Predecessor	Overlap (month)
A1 Cycle 1	2	-	-
A2 Analysis & Design	$\frac{3}{4}$	-	-
A3 Coding	1	A2	$\frac{1}{4}$
A4 Testing	$\frac{1}{2}$	A3	$\frac{1}{4}$
A5 Deployment & Training	$\frac{1}{4}$	A4	-
B1 Cycle 2	2	A1	1 $\frac{1}{4}$
B2 Analysis & Design	$\frac{1}{2}$	A2	-
B3 Coding	$\frac{3}{4}$	B2	-
B4 Testing	$\frac{3}{4}$	B3	$\frac{1}{4}$
B5 Deployment & Training	$\frac{1}{4}$	B4	-
C1 Cycle 3	1 $\frac{1}{4}$	B1	1
C2 Analysis & Design	$\frac{1}{4}$	B3	$\frac{1}{4}$
C3 Coding & Testing	$\frac{3}{4}$	C2	-
C4 Deployment & Training	$\frac{1}{2}$	C3	$\frac{1}{4}$
D1 Cycle 4			
D2 Analysis & Design			
D3 Coding & Testing			
D4 Deployment & Training			

- b) Analyse the Gantt Chart in Question 2a, which testing technique must be carried out in the software development for Increment 2, 3 and 4? Identify and design **TWO (2)** testing techniques with **ONE (1)** test case for EACH testing technique. You shall use the following format to design your test case.

(8 marks)

Program Name:					
Test Date: -					
No.	Test Case	Test Data	Expected Results	Actual Results	Remarks
				-	-

- c) Suggest and explain **ONE (1)** control modelling for CBB ERP system.

(2 marks)

- d) What could possibly motivate your *software engineers* throughout the whole project? Discuss **TWO (2)** motivational elements based on Maslow's Motivation Model by providing appropriate example for each element.

(4 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 3**

Please answer the following questions based on **CBB case study in Question 1**.

- a) Identify and explain a simple write up on: **TWO (2)** Process/Organizational Requirements, and **TWO (2)** External Requirements for CBB ERP system. (8 marks)
- b) What are the main concerns in User Interface (UI) design from the suppliers and sellers' perspective? Analyze **TWO (2)** main UI rules that must be fulfilled by this online ERP system to ease both the suppliers and sellers. Explain the rules by giving **ONE (1)** appropriate example for EACH rule. (8 marks)
- c) A *software engineer*, Eddie, is assigned to validate the System Requirements Specification (SRS). He found that there are two functional requirements which are conflict with each other. Besides that, Eddie also found that there are three missing requirements which he remembered during requirement analysis and elicitation stage. Nevertheless, there are two requirements which should not be included in SRS as these are additional functions. What are the **THREE (3)** aspects of requirements that have been checked by Eddie? Explain your answer. (9 marks)

[Total: 25 marks]

Question 4

Please answer the following questions based on **CBB case study in Question 1**.

- a) Design **TWO (2)** sets of stimuli-response for CBB ERP system for sellers in the process of selecting products from China suppliers and upload automatically into sellers' Malaysia store. (4 marks)
- b) Identify and explain **TWO (2)** periodic stimuli in CBB warehouse system. (4 marks)
- c) What are the factors that could affect CBB warehouse system's maintenance cost? Analyze and justify **THREE (3)** factors that might cause CBB warehouse system's high maintenance cost. Explain **ONE (1)** re-engineering activity to improve the CBB warehouse system in order to decrease the maintenance cost. (9 marks)
- d) Assuming that you are adapting *Aspect Oriented Software Engineering*, construct a viewpoint hierarchy with at least **FOUR (4)** viewpoints. Based on the viewpoint hierarchy, identify **ONE (1)** main concern for each viewpoint. (8 marks)

[Total: 25 marks]